Automatic Water Level Controller System

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Introduction

This project is based on building an Automatic Water Level Controller System using Arduino. The idea behind the project is to maintain the water level in a tank without needing manual monitoring. Water wastage and tank overflow are common problems in homes and buildings, so this system helps manage water more efficiently. The controller senses the level of water and automatically turns the motor ON or OFF, making the whole process reliable and convenient.

Problem Statement

Many households' face water wastage because tanks overflow when the motor is left running. Manual monitoring is time-consuming and sometimes ignored, leading to unnecessary power usage and water loss. To solve this issue, we need a system that detects the water level and controls the motor automatically. This project focuses on creating an affordable and simple water-level–based control system using Arduino.

Objectives

- To design an automatic circuit that monitors water levels using sensors.
- To implement a control mechanism that switches the motor ON/OFF automatically.
- To display water levels using LEDs for easy monitoring.
- To test and verify the correct functioning of sensors, relay, and motor.
- To develop a low-cost and reliable water management system.

Tools

- Arduino UNO – Used as the main controller to read sensor data and operate the relay.
- Water Level Sensors – Detect the current level of water in the tank.
- Relay Module – Controls the motor safely using Arduino signals.

Arduino UNO

• LEDs – Indicate LOW, MEDIUM, and FULL levels.

• Tinker cad – Used to design and test the circuit virtually before making the hardware.

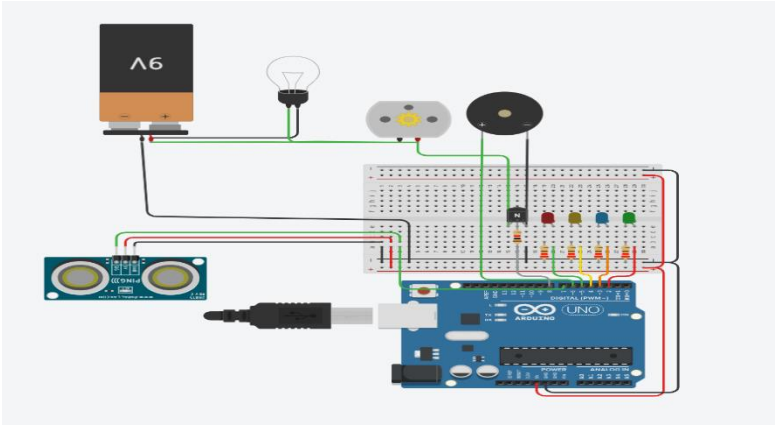
• Arduino IDE – Used to write and upload the program into the Arduino board.

Technique

- Simulation

Hardware Design

The system consists of a water sensor connected to Arduino, which reads the water level and processes the data. Based on the reading, Arduino sends a signal to the relay, which turns the motor ON when water is low and OFF when the tank becomes full. LEDs are also connected to show the tank status visually.



Main Components in the Design:

• Water Level Sensors

• Arduino UNO

• 5V Relay

• Motor (pump)

• LEDs for three water levels.

Ultrasonic-HC-Sr-04

LEDs

DC Motor

Implementation

1. First, the working of sensors, relay, and motor was studied.

2. The complete circuit was prepared on Tinker cad to verify the connections.

3. Arduino code was written in the IDE to check the water levels and control the motor.

4. All components were wired on a breadboard — sensors to analog pins, LEDs to digital pins, and relay to the control pin.

5. The code was uploaded to Arduino, and the system was tested by changing water levels manually.

6. Errors in wiring and threshold values were fixed during testing to achieve smooth operation.

Sample Arduino Logic Used:

• If level = LOW → Motor ON

• If level = MID → Motor remains in previous state

• If level = FULL → Motor OFF

Results and Discussion

The Automatic Water Level Controller worked successfully.

• When the tank reached LOW level, the motor turned ON immediately.

• LEDs displayed the real-time level clearly (LOW, MID, FULL).

• When the tank became FULL, the relay switched OFF the motor, preventing overflow.

Challenges and Limitations

• Sensor values fluctuated due to noise and required calibration.

• Breadboard wires sometimes became loose during testing.

• Relay switching created minor delays while turning the pump ON/OFF.

• The system's accuracy depends on proper sensor placement in the tank.

• High-power AC motors require additional safety precautions.

Conclusion

In this project, an automatic water level controller was successfully designed and implemented. The system accurately monitored water levels and controlled the pump without manual effort. It reduced water wastage, prevented overflow, and demonstrated the use of sensors and relays with Arduino. Through this project, we learned about integrating hardware components, troubleshooting circuits, and writing control logic. The final system worked reliably and achieved all project objectives.

Future Enhancements

- Add a mobile notification or IoT monitoring feature.
- Use an ultrasonic sensor for more accurate readings.
- Add buzzer alerts for critical water levels.
- Replace the breadboard with a PCB for permanent installation.